

THE UNIVERSITY OF DODOMA
COLLEGE OF INFORMATICS AND VIRTUAL EDUCATION
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
TEST TWO (60 MIN)

INTERNET PROGRAMMING AND APPLICATION I

Question One (25 Marks)

A web page is designed to manage a list of students. The page contains a form with the following elements:

- A text input field: `<input type="text" id="studentName" name="studentName" placeholder="Enter student name">`
- A submit button: `<button type="submit" id="addStudentBtn">Add Student</button>`
- An unordered list for displaying student names: `<ul id="studentList"></ul>`
- A paragraph for displaying messages: `<p id="message"></p>`

Write a jQuery script that performs the following tasks. When the form is submitted, prevent the default form submission behavior so that the page does not reload. Validate the text input field and check whether the student's name has been entered. If the input field is empty, display the message **"Student name is required"** in red inside the paragraph element. If a valid student name is entered, add the student's name as a new `<li>` element within the unordered list, clear the input field, and display the message **"Student added successfully"** in green. Furthermore, when a student name in the list is clicked, remove the corresponding list item from the list. Your solution should demonstrate the use of jQuery selectors, event handling, form validation, DOM manipulation, and CSS property manipulation.

Question two (15 Marks)

A hospital patient registration form has been built using HTML. The form contains an id of `"#patientForm"` and includes the following fields: Full Name (`#fullName`), Date of Birth (`#dob`), Phone Number (`#phone`), and Ward Selection (with `id="#ward"`).

There is also a read-only `#summary` div that should update in real time as the user types, and a pop-up that appears when the form is submitted successfully.

Using jQuery, write the code that:

- Validates the form on submission: all fields are required; the phone number must contain exactly 10 digits; the Full Name must be at least 3 characters. If any validation fails, display an appropriate error message next to the relevant field.
- Updates the `#summary` div live as the user types in the Full Name field, displaying: "Patient: [name] | Ward: [selected ward]".
- On successful validation, prevents the default form submission, then programmatically shows the pop-up to confirm submission and resets the form.